# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **Metals and Non-metals** | Class 10 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** Both A and R are true and R is the correct explanation of A. | Kerosene prevents contact with air/water. |
| 2 | **(a)** Y | Reacting with cold water indicates more reactivity. |
| 3 | **(c)** $CuO$ and $CO_2$ | Calcination: $CuCO_3 \longrightarrow CuO + CO_2$. |
| 4 | **(a)** Both A and R are true and R is the correct explanation of A. | Oxide layer protects the metal. |
| 5 | **(c)** S | Each displaces the next, so S is lowest. |
| 6 | **(d)** copper | $CuO + H_2 \longrightarrow Cu + H_2O$. |
| 7 | **(a)** Both A and R are true and R is the correct explanation of A. | Layers repel and crystal breaks. |
| 8 | **(a)** $FeSO_4$ and $CuSO_4$ | Zn is more reactive than Fe and Cu but less than Mg. |
| 9 | **(a)** copper | Cu and CuO. |
| 10 | **(a)** Both A and R are true and R is the correct explanation of A. | Delocalised electrons carry the current. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Oxide is amphoteric (1); Zinc or aluminium (1). | 1+1 |
| 12 | Reduction of oxide is easier by carbon (1); So sulphides/carbonates are first converted to oxides (1). | 1+1 |
| 13 | Protective $Al_2O_3$ film (1); Film prevents contact with air/water (1). | 1+1 |
| 14 | Very exothermic, hydrogen catches fire (1); $2\,Na + 2\,H_2O \longrightarrow 2\,NaOH + H_2$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $ZnO$ is the oxidising agent and C is the reducing agent (1); Redox, reduction of metal oxide with carbon (1); Zinc is moderately reactive, so carbon can reduce its oxide (1). | 1+1+1 |
| 16 | Salt water increases electrical conductivity (1); Faster electrochemical corrosion (1); Rust formation speeds up (1). | 1+1+1 |
| 17 | NaCl gives $Na^+$ and $Cl^-$ ions (1); Sugar remains as neutral molecules (1); No ions in sugar solution (1). | 1+1+1 |
| | **OR:** A, B more reactive than Fe (1); C is between Fe and Cu (1); Order: (A, B) > Fe > C > Cu > D (1). | 1+1+1 |
| 18 | X is $CuO$, Y is copper (1); $CuO + H_2 \longrightarrow Cu + H_2O$ (1); $2\,Cu + O_2 \longrightarrow 2\,CuO$ (1). | 1+1+1 |
| | **OR:** Oxides of Na, K, Ca dissolve (1); CuO is insoluble (1); Soluble ones form alkalis (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Order: A > B > C > D (1); A: electrolysis of molten salt (very reactive) (1); D: occurs free or by heating its ore (1); Example set: Mg, Zn, Cu, Ag (1); Reason: a more reactive metal displaces a less reactive one (1). | 1+1+1+1+1 |
| | **OR:** (a) Reacts with air and water (1); (b) Malleable, forms $Al_2O_3$ layer (1); (c) Pure gold is soft (1); (d) Like ions repel when layers shift (1); Each reason relates to property (1). | 1+1+1+1+1 |
| 20 | Purification of bauxite to $Al_2O_3$ (1); Molten alumina with cryolite (1); Cathode: $Al^{3+} + 3\,e^- \longrightarrow Al$ (1); Anode: $2\,O^{2-} \longrightarrow O_2 + 4\,e^-$; graphite burns (1); Cryolite lowers the temperature and raises conductivity (1). | 1+1+1+1+1 |
| | **OR:** Black layer turns brown (1); $CuO + H_2 \longrightarrow Cu + H_2O$ (1); $CuO$ is the oxidising agent (1); $H_2$ is the reducing agent (1); Copper is regenerated (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

## ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **Metals and Non-metals** | Class 10 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** $Na_2O$ | Soluble basic oxides form alkalis. |
| 2 | **(b)** turns red litmus blue | NaOH is formed. |
| 3 | **(a)** Both A and R are true and R is the correct explanation of A. | Sacrificial protection. |
| 4 | **(a)** $MCl_3$ | $M^{3+}$. |
| 5 | **(d)** aluminium | Cryolite lowers its melting point. |
| 6 | **(a)** Both A and R are true and R is the correct explanation of A. | Reactivity decides displacement. |
| 7 | **(d)** 31.75 g | 0.5 x 63.5 = 31.75 g. |
| 8 | **(b)** tin and lead | Solder-type alloy. |
| 9 | **(a)** Both A and R are true and R is the correct explanation of A. | Carbon cannot reduce oxide of a metal above it. |
| 10 | **(a)** 112 g | 2 mol Al gives 2 mol Fe. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | Silver is below hydrogen (1); Sodium or calcium displaces hydrogen from water (1). | 1+1 |
| 12 | Strong electrostatic forces (1); Displacement of layers brings like ions close, causing repulsion (1). | 1+1 |
| 13 | $HNO_3$ is a strong oxidising agent (1); It oxidises $H_2$ to water and itself forms nitrogen oxides (1). | 1+1 |
| 14 | Zinc is more reactive than tin and dissolves in food acid (1); Zinc salts are harmful (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Copper reacts with $CO_2$ and moisture forming basic copper carbonate (1); Acids in lemon/tamarind react with the green layer (1); The basic carbonate dissolves leaving shiny copper (1). | 1+1+1 |
| 16 | Roasting in air gives copper(I) oxide (1); Reduction by self-reduction with $Cu_2S$ (1); Electrolytic refining gives pure copper (1). | 1+1+1 |
| 17 | Brass is harder (1); More corrosion-resistant (1); Easily cast and has golden colour (1). | 1+1+1 |
| | **OR:** X is sodium (1); $4\,Na + O_2 \longrightarrow 2\,Na_2O$ (1); $Na_2O + H_2O \longrightarrow 2\,NaOH$ (1). | 1+1+1 |
| 18 | Ag and Au are least reactive (1); They do not react with oxygen, water (1); Sodium is highly reactive and forms stable compounds (1). | 1+1+1 |
| | **OR:** Foil: malleability (1); Wires: conductivity and ductility (1); Aircraft: light weight when alloyed (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | $ZnO + C \longrightarrow Zn + CO$ (1); Molar mass ZnO = 81 g (1); 162 g = 2 mol (1); Zinc = 130 g (1); CO = 2 mol = 44.8 L (1). | 1+1+1+1+1 |
| | **OR:** NaCl (molten): Na at cathode, $Cl_2$ at anode (1); Copper: impure anode, pure cathode (1); NaCl: extraction; Cu: refining (1); Both need electric current and a cathode (1); $Cu^{2+}$ ions in electrolyte (1). | 1+1+1+1+1 |
| 20 | X: 2,8,8,2; Cl: 2,8,7 (1); X loses 2 electrons, each Cl gains 1 (1); Formula $XCl_2$ = $CaCl_2$ (1); Ionic, high melting point (1); Soluble in water, conducts when molten (1). | 1+1+1+1+1 |
| | **OR:** Ore is a profitable mineral (1); Roasting in excess air, calcination in limited air (1); Amalgam contains mercury (1); Galvanisation prevents corrosion with zinc (1); Ionic: transfer of electrons, covalent: sharing (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

## ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **Metals and Non-metals** | Class 10 Science | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** zinc blocks on ship hull | Zinc corrodes first. |
| 2 | **(a)** Both A and R are true and R is the correct explanation of A. | Delocalised electrons exist. |
| 3 | **(c)** charge +2, formula MgO | Mg (2,8,2) loses 2 electrons. |
| 4 | **(c)** roasting -> reduction -> refining | Sulphide ores are roasted first. |
| 5 | **(c)** A is true but R is false. | Calcium is denser; hydrogen bubbles make it float. R false. |
| 6 | **(c)** $M_2S_3$ | $M^{3+}$ and $S^{2-}$: 2 x 3 = 3 x 2. |
| 7 | **(a)** a sulphide | Sulphide ores need excess air. |
| 8 | **(a)** Both A and R are true and R is the correct explanation of A. | Alloyed for hardness. |
| 9 | **(c)** hydrogen sulphide | It forms silver sulphide. |
| 10 | **(a)** are less reactive than copper and do not dissolve | They remain undissolved. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | M is magnesium (1); $2\,Mg + O_2 \longrightarrow 2\,MgO$ (1). | 1+1 |
| 12 | $Fe_2O_3 + 2\,Al \longrightarrow Al_2O_3 + 2\,Fe$: 2 mol Al gives 2 mol Fe (1); Mass of Fe = 112 g (1). | 1+1 |
| 13 | Anode: chlorine (1); Cathode: sodium (1). | 1+1 |
| 14 | (i) Yes, Cu is more reactive than Ag (1); (ii) No, Ag is less reactive (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | Fe: reduction with carbon/CO, medium reactivity (1); Na: electrolysis of molten NaCl, high reactivity (1); Carbon cannot reduce $Na^+$ (1). | 1+1+1 |
| 16 | Mixture shows properties of both (1); Heating gives a compound FeS (1); Magnet does not attract it (1). | 1+1+1 |
| 17 | $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$: 65 g gives 63.5 g (1); 13 g is 0.2 mol (1); Copper = 12.7 g (1). | 1+1+1 |
| | **OR:** Alumina melts at very high temperature (1); Cryolite lowers the melting point and increases conductivity (1); Graphite electrodes carry current (1). | 1+1+1 |
| 18 | Anodising: make Al the anode in dilute $H_2SO_4$ (1); $Al_2O_3$ layer becomes thicker (1); It resists corrosion and can be dyed (1). | 1+1+1 |
| | **OR:** Highly reactive: electrolysis (1); Moderately reactive: reduction by carbon (1); Least reactive: heating or found free (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Na/K react with all, even cold water (1); Mg, Al, Zn, Fe: acids and steam (1); Cu: reacts with oxygen only on heating (1); Ag, Au: no reaction (1); Displacement: upper metal displaces lower (1). | 1+1+1+1+1 |
| | **OR:** M could be iron (1); Reacts with dilute HCl: $Fe + 2\,HCl \longrightarrow FeCl_2 + H_2$ (1); With steam: $3\,Fe + 4\,H_2O \longrightarrow Fe_3O_4 + 4\,H_2$ (1); With $CuSO_4$: displaces Cu (1); Does not displace Zn from $ZnSO_4$ (1). | 1+1+1+1+1 |
| 20 | (a) no rust, no water (1); (b) no rust, no oxygen (1); (c) rust (1); Rusting needs both oxygen and moisture (1); Prevention: paint, oil, galvanising, alloying (1). | 1+1+1+1+1 |
| | **OR:** Reactive metals form stable compounds (1); $ZnCO_3 \longrightarrow ZnO + CO_2$ (calcination) (1); $2\,ZnS + 3\,O_2 \longrightarrow 2\,ZnO + 2\,SO_2$ (roasting) (1); $ZnO + C \longrightarrow Zn + CO$ (1); Roasting/calcination makes oxide which is reduced (1). | 1+1+1+1+1 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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